

Malware Analysis Report For MachO-OSX-x86-ls

Binary Type

Mach-O (Mach Object). (Approved)

Analyzing File

- **File Name:** MachO-OSX-x86-ls
 - **File Extension:**
 - **File Type:** Mach-O (Mach Object).
 - **MD5 Hash:** df2580eaf51e15e23de3db979992af1e
 - **SHA256 Hash:** f6192f91dbf2bb98d846a0e409bcd2895cb8e04016f0476c3c4d22af3713f5e0
-

File Identification

File detected as a valid macOS/iOS executable.

Mach-O File Analysis

Mach-O File Analysis: True

Mach-O File Imports

No imported functions were found.

Mach-O File Exports

No exported functions were found.

String Analysis

ASCII Strings

- __PAGEZERO
- __TEXT
- __text
- __TEXT
- _symbolstub
- __TEXT
- _stubhelper
- __TEXT
- __cstring
- __TEXT
- _unwindinfo
- __TEXT
- __DATA
- _programvars

- __DATA
- _n/symbol_ptr
- __DATA
- _/asymbol_ptr
- __DATA
- __data
- __DATA
- __const
- __DATA
- __common
- __DATA
- __DATA
- __LINKEDIT
- /usr/lib/dyld
- /usr/lib/libncurses.5.4.dylib
- /usr/lib/libutil.dylib
- /usr/lib/libSystem.B.dylib
- \$FreeBSD: src/bin/ls/cmp.c,v 1.12 2002/06/30 05:13:54 obrien Exp \$
- LS_COLWIDTHS
- malloc
- %llu : %qu : %lu : %i : %i : %i : %qu : %lu : %lu
- %s: %s
- fflagstostr
- fts_open
- %s: directory causes a cycle
- bin/ls
- fts_read
- COLUMNS
- CLICOLOR
- 1@ABCFGHLOPRSTUWabcdefghijklmnopqrstuvwxyz
- CLICOLOR_FORCE
- LSCOLORS
- \$FreeBSD: src/bin/ls/ls.c,v 1.66 2002/09/21 01:28:36 wollman Exp \$
- exfxcxdxbxegedabagacad
- warn: LSCOLORS should use characters a-h instead of 0-9 (see the manual page)
- error: invalid character '%c' in LSCOLORS env var
- %*llu
- Unix2003
- total %qu
- %e %b %T %Y
- %b %e %T %Y
- %e %b %R
- %b %e %R
- %e %b %Y
- %b %e %Y
- ls: %s: %s
- Unable to translate qualifier on ACL
- unknown
- inherited
- %d: %s%s %s
- add_file
- execute

- search
- delete
- append
- add_subdirectory
- delete_child
- readattr
- writeattr
- readextattr
- writeextattr
- readsecurity
- writesecurity
- file_inherit
- directory_inherit
- limit_inherit
- only_inherit
- %s%s %*u
- %s%s %u %-s
- %s%s %u %-s %-*s
- %3d, 0x%08x
- %3d, %3d
- %s%qu
- \$FreeBSD: src/bin/ls/print.c,v 1.57 2002/08/29 14:29:09 keramida Exp \$
- usage: ls [-ABCFGHLOPRSTUWabcdeghiklmnopqrstuw1] [file ...]
- \$FreeBSD: src/bin/ls/util.c,v 1.38 2005/06/03 11:05:58 dd Exp \$
- UBXA\Bp
- @__DefaultRuneLocale
- @_stackchkguard
- @_stderr
- @_stdout
- @_optind
- @dyldstubfinder
- @_divdi3
- @_error
- @_maskrune
- @_moddi3
- @_snprintf_chk
- @_stackchkfail
- @_tolower
- @_udivdi3
- @_umoddi3
- @aclfree
- @aclget_entry
- @aclgetflagnp
- @aclgetflagsetnp
- @aclgetlinknp
- @aclgetpermnp
- @aclget_permset
- @aclget_qualifier
- @aclgettagtype
- @_atoi
- @compatmode

- `@_exit`
- `@_flagstotr`
- `@_fprintf`
- `@_fputs$UNIX2003`
- `@_free`
- `@ftschildren$INODE64`
- `@ftsclose$INODE64`
- `@ftsopen$INODE64`
- `@ftsread$INODE64`
- `@ftsset$INODE64`
- `@_getbsize`
- `@_getenv`
- `@_getgrgid`
- `@_getopt$UNIX2003`
- `@_getpid`
- `@_getpwuid`
- `@_getuid`
- `@_getxattr`
- `@groupfrom_gid`
- `@humanizenumbers`
- `@_ioctl`
- `@_isatty`
- `@_kill$UNIX2003`
- `@_listxattr`
- `@_localtime`
- `@_malloc`
- `@mbuuidtoid`
- `@mbuuidtostring`
- `@_mbrtowc`
- `@_memchr`
- `@nlanginfo`
- `@_printf`
- `@_putchar`
- `@_readlink`
- `@_realloc`
- `@_setenv$UNIX2003`
- `@_setlocale`
- `@_signal`
- `@_snprintf`
- `@_sscanf`
- `@_strcoll`
- `@_strcpy`
- `@_strdup`
- `@_strerror$UNIX2003`
- `@_strftime$UNIX2003`
- `@_strlen`
- `@_strmode`
- `@_tgetent`
- `@_tgetstr`
- `@_tgoto`
- `@_time`
- `@_tputs`

- @userfrom_uid
- @_warn
- @_warnx
- @_wewidth
- @_write\$UNIX2003
- _mhexecute_header
- >##YrYYYYrYYY
- _mhexecute_header
- __DefaultRuneLocale
- _divdi3
- _error
- _maskrune
- _moddi3
- _snprintf_chk
- _stackchkfail
- _stackchkguard
- _stderrp
- _stdoutp
- _tolower
- _udivdi3
- _umoddi3
- ac/free
- ac/get_entry
- ac/getflagnp
- ac/getflagsetnp
- ac/getlinknp
- ac/getpermpnp
- ac/get_permset
- ac/get_qualifier
- ac/gettagtype
- compatmode
- _fflagstostr
- _fprintf
- _fputs\$UNIX2003
- ftschildren\$INODE64
- ftsclose\$INODE64
- ftsopen\$INODE64
- ftsread\$INODE64
- ftsset\$INODE64
- _getbssize
- _getenv
- _getgrgid
- _getopt\$UNIX2003
- _getpid
- _getpwuid
- _getuid
- _getxattr
- groupfrom_gid
- humanizenumber
- _ioctl
- _isatty

- `_kill$UNIX2003`
- `_listxattr`
- `_localtime`
- `_malloc`
- `mbnuidtoid`
- `mbnuidtostring`
- `_mbrtowc`
- `_memchr`
- `nllanginfo`
- `_optind`
- `_printf`
- `_putchar`
- `_readlink`
- `_realloc`
- `_setenv$UNIX2003`
- `_setlocale`
- `_signal`
- `_snprintf`
- `_sscanf`
- `_strcoll`
- `_strcpy`
- `_strdup`
- `_strerror$UNIX2003`
- `_strftime$UNIX2003`
- `_strlen`
- `_strmode`
- `_tgetent`
- `_tgetstr`
- `_tgoto`
- `_tputs`
- `userfrom_uid`
- `_warnx`
- `_wcurwidth`
- `_write$UNIX2003`
- `dyldstubbinder`
- `radr://5614542`
- `com.apple.ls`
- `libncurses.5`
- `com.apple.libutil1`
- `libSystem.B`
- `Apple Inc.1&0$`
- `Apple Certification Authority1`
- `Apple Root CA0`
- `070214211919Z`
- `150214211919Z0`
- `Apple Inc.1&0$`
- `Apple Certification Authority1301`
- `*Apple Code Signing Certification Authority0`
- `drmljGa`
- `%http://www.apple.com/appleca/root.crl0`
- `QKK:laN`
- `Apple Inc.1&0$`

- Apple Certification Authority1
- Apple Root CA0
- 060425214036Z
- 350209214036Z0b1
- Apple Inc.1&0\$
- Apple Certification Authority1
- Apple Root CA0
- https://www.apple.com/appleca/0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority0
- 070223220256Z
- 150114220256Z0V1
- Apple Inc.1
- Apple Software1
- Software Signing0
- EA3%|k=
- http://www.apple.com/appleca/0
- Reliance on this certificate by any party assumes acceptance of the then applicable standard terms a
- ,http://www.apple.com/appleca/codesigning.crl0
- Apple Inc.1&0\$
- Apple Certification Authority1301
- *Apple Code Signing Certification Authority
- CHwP"~K&(

UTF-8 Strings

No detected.

UTF-16LE Strings

No detected.

Extracted URLs

- http://www.apple.com/appleca/root.crl0
- https://www.apple.com/appleca/0
- http://www.apple.com/appleca/0
- http://www.apple.com/appleca/codesigning.crl0

Base64 Strings

No valid Base64-encoded strings found.

File Entropy Analysis

- **Entropy:** 5.58
 - **Status:** This file likely contains Standard Text or human-readable text.
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VirusTotal Analysis

Scan Date: 2023-02-06 20:25:08

Scan Results

- Malicious: 0
- Suspicious: 0
- Undetected: 62
- Harmless: 0
- Timeout: 0
- Confirmed-timeout: 0
- Failure: 0
- Type-unsupported: 12

Community Votes

- Harmless: 0
- Malicious: 0

Detailed Engine Results

Engine Name	Detection Category	Detection
No results	No results	No results
